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DSMC SIMULATIONS OF SQUEEZE FILM BETWEEN A MICRO BEAM UNDERGOING LARGE AMPLITUDE OSCILLATIONS AND A SUBSTRATE

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ABSTRACT

This paper investigates the behavior of a gas film in a micro RF switch. A Two-dimensional numerical study of the flow field is performed as the micro-beam oscillates harmonically between its equilibrium position and the fixed substrate underneath. Unlike previous work in literature, the beam undergoes large displacements throughout the film gap thickness and the behavior of the gas film along with its impact on the moving RF switch (force exerted by gas on the beam's front and back faces) are discussed. Since the gas film thickness is of the order of few microns (i.e. $0.01 < Kn < 1$), the rarefied gas exists in the non-continuum regime and, as such, the Direct Simulation Monte Carlo (DSMC) method is used to simulate the fluid behavior. The impact of the squeeze film on the beam is investigated over a range of frequencies, velocity amplitudes, and for different film gases, corresponding to ranges of dimensionless flow parameters such as the Reynolds (Re), Strouhal (St) and Mach (Ma) numbers on the gas film behavior.

NOMENCLATURE

DSMC Direct Simulation Monte Carlo.
 b Micro-Beam width.
 V_a Velocity amplitude.
 f Oscillation frequency.

M_d Molecular diameter.
 M_m Molecular mass.
 γ Specific heat ratio.
 ρ Density.
 λ Mean free path.
 a Speed of sound.
 μ Dynamic viscosity.
 k Boltzmann constant.
 h_0 Gap film thickness.
 Kn Knudsen number.
 Re Reynolds number.
 St Strouhal number.
 Ma Mach number.
 b^* Normalized beam width.
 τ_{conv} Convection time scale.
 τ_{diff} Diffusion time scale.
 τ_{osc} Oscillation time scale.
 τ_{coll} Molecular collision time scale.
 T_0 Ambient temperature.
 P_0 Ambient pressure.
 σ_d Squeeze number.

INTRODUCTION

With the rapid advancements in MEMS technologies in the early 90's, there emerged a need to better understand the role of squeeze films and their impact on moving micro-structures.

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Applications include RF microswitches, micromirrors for optical switching, MEMS accelerometers and gyroscopes, micromotors and micro gas turbines ([1], [2], [3], [4], [5], [6]). A starting point for researchers is Reynolds lubrication theorem, which was first derived by Blech [7], due to the interest in fluid film behavior in bearings. To better understand the behavior of a MEMS RF switch, many models have been suggested in literature to describe the physical behavior of the gas (usually air) film between the vibrating structure and the underneath fixed substrate. Andrews et al. [8] conducted an experiment to compare the vibrating plate behavior and gas film flow coefficients (spring and damper) with the theoretical results of [7] and discussed the accuracy of the model at different frequency ranges and plate-structure spacings. Zook et al. [9] experimentally measured the quality factor (i.e. the overall change of energy for molecules hitting the microbeam within one period of oscillation) of a vibrating double-clamped beam microsensor and their results have been widely used as a benchmark for many models in literature. Guo and Alexeenko [10] used the ellipsoidal statistical Bhatnagar-Gross-Krook model of the equilibrium distribution function in the quasi-steady Boltzmann kinetic model to derive expressions for the gas film damping coefficient and the quality factor. Bao et al. [11] proposed a new model for air damping coefficients in vibratory micro-sensors operating under low ambient pressure. Their model uses energy transfer between the moving molecules and the vibrating plate and therefore surpasses Christian's model [12] in two main aspects: the new model is a function of the structure geometry and takes into account the presence of boundaries near the vibrating structure. The effect of the presence of a fixed boundary near a vibrating substrate was discussed by Newell [13] who studied its effect on the damping coefficient of the squeezed gas film (i.e. quality factor). Hutcherson and Ye [14] modeled the quality factor of a vibrating micro-beam and discussed the validity of previous models in literature depending on the boundary conditions assumed (i.e. free oscillating structure as compared to a structure oscillating near a fixed wall). They also relaxed the constant velocity assumption used in Bao's model [11] and the simulated quality factor values for the microbeam resonator were in excellent agreement with experimental results especially in the free-molecular regime as compared to the transition regime since their MD code was based on the absence of intermolecular forces among the gas molecules. Li and Fang [15] extended the work of Hutcherson and Ye that was limited to parallel plate vibrations to account for rotating plates and bending of micro-beams. Bao et al [16] also studied the effect of beam tilting by using the non-linear Reynolds equation and analytically solving for the coefficient of damping torque at non-uniform gap distance. The main restriction of their model is that the damping pressure should be less than the ambient pressure leading to a small squeeze number with the air film viscous force dominant over its spring force. It should be pointed out that the latter assumption is only true

for small squeeze number where the operating frequency is significantly less than the mirror first natural frequency leading to an "incompressible" behavior of the gas film as explained by Langlois [17] and Starr [18]. Veijola et al [19] used analytical models for the gas flow damping coefficient to solve for the capacitance of a perforated plate RF MEMS switch undergoing small amplitude vibrations using circuit analysis. The validity of their model is restricted to relatively low switching frequencies (and consequently Reynolds number) and inertia free flows. They modeled the gas flow damping coefficient as sum of contributions of three regions: between substrate and moving plate, through perforated holes, and in the intermediate region between the first two. Veijola [20] extended the Reynolds squeezed film damping by introducing gas rarefaction and inertia effects and used reduced order modeling (circuit analysis) to solve the resulting improved equations. Gallis and Torczynski [21] modified the Reynolds equation model for squeezed film damping by correcting for the pressure boundary conditions at the beam tips compared to the trivial ambient pressure conditions used before. They used the Navier-Stokes slip-jump (NSSJ) and the Direct Simulation Monte Carlo (DSMC) methods to determine the new pressure boundary conditions coefficients. J W Lee et al [22] extended the work of [20] and [21] by accounting for the higher order bending modes to derive a new squeezed film damping model. Such an extension provided a improved model that yielded better results when compared to experimental data carried by the authors themselves since it accounts for the bending of the beam structure. Sumali [23] conducted an experiment using a laser doppler vibrometer to find the gas film damping ratio for a vibrating micro parallel plate structure in the free-molecular regime and compared his results to various models in literature ([7], [8], [11], [19], [20], [21]) over a range of Knudsen number values. Mol et al [24] argued that Sumali's [23] experimental method is based on modal analysis (i.e. frequency domain) and is therefore complicated since it needs a predefined input driving excitation force. Alternatively, they proposed a time domain electrostatically actuated experiment and validated the results of ([11], [19], [20]).

The objective of this work is to use the Direct Simulation Monte Carlo (DSMC) ([25], [26], and [27]) method to model the behavior of the gas film between a micro-beam (RF switch) undergoing large amplitude oscillations and the fixed substrate underneath. Figure 1 shows a flow chart of the basic steps involved for the DSMC algorithm as implemented in this work to model the non-continuum gas flow behaviour around an RF micro switch cross-section as shown in Fig. 2. Figure 3 shows the two dimensional computational domain as modeled in the DSMC code defined by the rectangle ABCD due to symmetry about the beam's centerline ZZ' (Fig. 2). AB is defined as a symmetry line undergoing specular reflection whereas the substrate BC along with the beam faces (EF, FG, and GH) are all modeled as diffusive walls at ambient temperature. The enclosed region

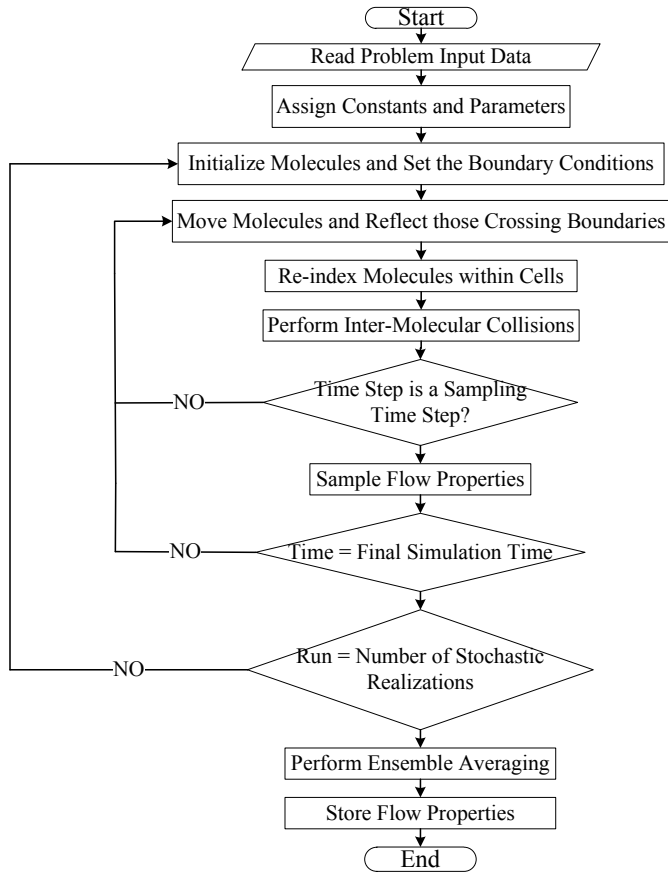


FIGURE 1. FLOW CHART OF THE DSMC ALGORITHM.

$ADCC'IA'A$ contains the reservoir cells that are modeled to impose the desired boundary conditions of ambient temperature and pressure.

Unlike previous works in the literature that are limited to small vibration amplitudes or frequency ranges where the force exerted by the gas on the beam is modeled without detailed investigation of the gas behavior, in this work, the beam is allowed to move at any desired velocity amplitude and frequency (i.e. the distance covered by the vibrating beam could extended over most of the film gap) and the gas behavior is modeled by the interaction of molecules among themselves and the oscillating micro-beam. Therefore, all thermodynamic properties (density, pressure, temperature, etc.) of the gas along with its velocity can be found at any position within the gap film. This way, a complete analysis of the gas behavior along with its effect on the vibrating beam can be performed which will, hopefully, improve micro RF switches design.

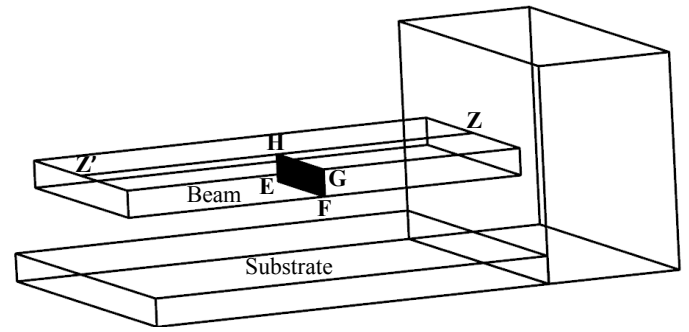


FIGURE 2. THREE DIMENSIONAL SCHEMATIC OF THE MICRO RF SWITCH.

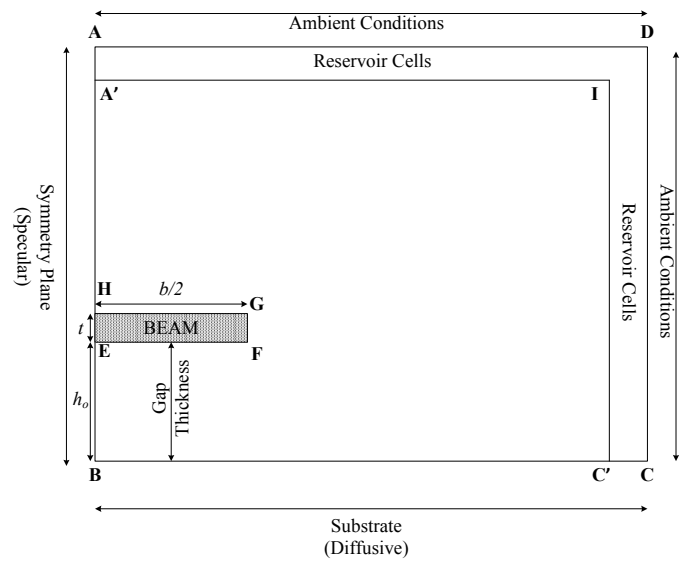


FIGURE 3. TWO DIMENSIONAL SIMULATION DOMAIN.

DIMENSIONLESS GROUPS

To better understand the behavior of the RF switch under different conditions, dimensionless groups are defined based on the physical parameters of interest. To start with the oscillating beam, the parameters involved are the beam width (b), beam velocity with amplitude (V_a) and direction normal to the fixed substrate, and the vibrating frequency (f). For the gas medium surrounding the RF switch, the physical parameters involved are the gas molecular diameter (M_d), the gas molecular mass (M_m), the gas specific heat ratio (γ), the gas density (ρ), the gas mean free path

$$\lambda = \frac{M_m}{\sqrt{2\pi\rho M_d^2}}, \quad (1)$$

the speed of sound in the gas medium

$$a = \sqrt{\frac{\gamma k T}{M_m}}, \quad (2)$$

where k is the Boltzmann constant and T is the temperature, and the gas dynamic viscosity

$$\mu = \frac{M_m}{3\sqrt{2}\pi M_d^2} \sqrt{\frac{8kT}{\pi M_m}}. \quad (3)$$

Since viscosity is a continuum concept and the gas flow in the gap film is in the non-continuum regime ($Kn > 0.01$), the effective viscosity (μ_{eff}) is used according to Veijola's model [28]:

$$\mu_{eff} = \frac{\mu}{1 + 9.638 \left(\frac{\lambda}{h_0}\right)^{1.159}} \quad (4)$$

This correction of the viscosity accounts for the boundary slip flow due to gas rarefaction near the wall.

For the simulation domain, the gap film thickness (h_0) is the main physical parameter influencing the gas flow behavior.

A set of dimensionless groups that capture all the physical parameters mentioned above are formed. The Knudsen number, which determines the degree of rarefaction and the flow regime (continuum vs non-continuum), is the ratio of the molecular mean free path to the characteristic flow length scale

$$Kn = \frac{\lambda}{h_0} \quad (5)$$

The Reynolds number is defined as the ratio of the momentum convection term (inertia forces) to the momentum diffusion term (viscous forces).

$$Re = \frac{V_a h_0}{\nu} = \frac{\tau_{diff}}{\tau_{conv}} \quad (6)$$

where the film gap thickness (h_0) is the characteristic length scale, the beam's velocity amplitude (V_a) is the characteristic speed scale, $\nu = \frac{\mu_{eff}}{\rho}$ is the kinematic viscosity, $\tau_{conv} = \frac{h_0}{V_a}$ and $\tau_{diff} = \frac{h_0^2}{\nu}$ are the convection and diffusion time scales respectively. Except for very large values of V_a , $Re \ll 1$ and consequently the convection component of the inertia term can be neglected. The RF switch undergoes harmonic oscillatory motion (unsteady), and the Strouhal number (St) is introduced to

non-dimensionalize the unsteady component of the inertia term. The Strouhal number is the ratio of inertial forces due to flow oscillations (i.e. local acceleration due to change of velocity with time where the oscillatory time scale is given by $\tau_{osc} = \frac{1}{f}$) to the convection term (i.e. acceleration due to spatial change of velocity)

$$St = \frac{\tau_{conv}}{\tau_{osc}} = \frac{f h_0}{V_a} = \frac{h_0}{2\pi\delta} \quad (7)$$

where (δ), the maximum displacement of the vibrating beam, is

$$\delta = \int_0^{\frac{1}{2f}} V_a \sin(2\pi f t) dt = \frac{V_a}{2\pi f} \quad (8)$$

Note that, alternatively, Strouhal number measures the ratio of the gap thickness to the maximum distance traversed by the vibrating beam. Another dimensionless group of interest is the Mach Number (Ma), the ratio of the characteristic gas flow speed to the speed of sound in the gas medium,

$$Ma = \frac{V_a}{a} \quad (9)$$

where (a), the speed of sound, is expressed as $a = \sqrt{\frac{\gamma k T}{M_m}}$. Finally, the beam width (b) is normalized according to the characteristic length scale (h_0) as

$$b^* = \frac{b}{h_0} \quad (10)$$

Note that the squeeze film number, which measures the relative importance of viscous forces to spring forces, may be expressed as:

$$\sigma_d = \frac{24\pi\mu b^2 f}{h_0^2 P_0} = \frac{24\pi\gamma St Ma^2 (b^*)^2}{Re} \quad (11)$$

Noting that a scales as the most probable speed, the ratios of the collision time scale to convection, diffusion, and oscillation time scales, expressed in terms of dimensionless quantities, are respectively

$$\frac{\tau_{coll}}{\tau_{conv}} \approx \frac{\lambda/a}{h_0/V_a} = (Ma)(Kn) \quad (12)$$

$$\frac{\tau_{coll}}{\tau_{diff}} = \frac{\tau_{coll}}{\tau_{conv}} \frac{\tau_{conv}}{\tau_{diff}} \approx \frac{(Ma)(Kn)}{Re} \quad (13)$$

$$\frac{\tau_{coll}}{\tau_{osc}} = \frac{\tau_{coll}}{\tau_{conv}} \frac{\tau_{conv}}{\tau_{osc}} \approx (Ma)(Kn)(St) \quad (14)$$

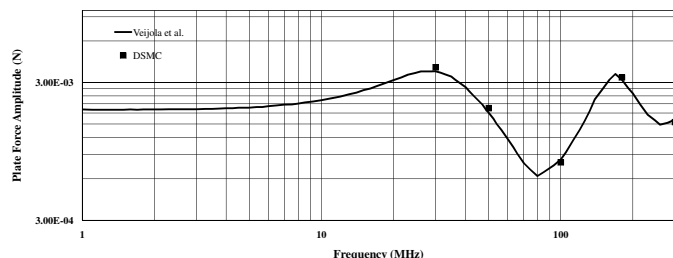


FIGURE 4. FORCE AMPLITUDE ON THE MICRO-BEAM'S FRONT FACE.

RESULTS AND DISCUSSION

Code Validation

The DSMC code is validated by for the case of a microbeam of length $5\mu\text{m}$ oscillating at a frequency of 1GHz and speed amplitude of 1m/s with a gap thickness of $1\mu\text{m}$ as studied by Veijola et al [29]. Under these conditions, Knudsen number is 0.0682 and the flow exists in the slip-flow regime. It should be pointed out that since the speed amplitude (1m/s) is much smaller than the average molecular speed at room temperature ($\sim 330\text{m/s}$), a relatively large number of molecules is used in the DSMC algorithm and values are averaged out over a large number of stochastic realizations to ensure minimal noise effects. The DSMC results, plotted in Fig. 4 for five different frequencies (30, 50, 100, 180, and 300 MHz), are in good agreement with those obtained by Veijola et al [29] using the finite element solver ELMER. Note that ELMER solves the linearized Navier-Stokes equation (i.e. Reynolds equation) and is thus restricted to small amplitude oscillations. In this paper, DSMC is employed to model a wide range of beam oscillation frequencies and amplitudes as discussed next.

Hereafter, simulations are performed that cover a range of values of the various dimensionless groups as listed in Tab. 1. All results are presented in terms of normalized quantities after simulating the flow over a large number of periods (i.e. quasi-steady state); the flow velocity normal to the beam is normalized with respect to the beam's velocity amplitude (V_a), pressure is normalized with respect to the relevant gas atmospheric pressure at 20°C , density is normalized with respect to relevant gas density at 20°C , force is normalized with respect to the product of the atmospheric pressure and the microbeam's area, and time is normalized (τ_{norm}) with respect to the beam's oscillation period (T). Table 2 summarizes the geometrical parameters used and the ambient gas medium physical conditions. The relation between the gas flow behavior and the force imparted on the beam is interpreted in the relative importance of diffusion and inertia, where the latter is the superposition of an unsteady term and a convective term. Note that the ratio of the convective inertia term to the diffusion term scales as Re , the ratio of the unsteady inertia term to the convective inertia term scales as St , and the ratio of

	Kn	Re ($\tau_{\text{diff}}/\tau_{\text{con}}$)	St ($\tau_{\text{con}}/\tau_{\text{osc}}$)	Ma	St*Ma ² /Re ($\sigma_d/(24\pi\eta b^2)$)	Re*St ($\tau_{\text{diff}}/\tau_{\text{osc}}$)	Ma*Kn ($\tau_{\text{con}}/\tau_{\text{con}}$)	Ma*Kn*St ($\tau_{\text{con}}/\tau_{\text{osc}}$)	Ma*Kn/Re ($\tau_{\text{con}}/\tau_{\text{diff}}$)
CASE 1: Varying V_a ($f=1$ GHz, Gas: air)									
V_a (m/s)									
20	0.068	1.889	50	0.058	0.09	94.43	0.004	0.198	0.002
100	0.068	9.443	10	0.292	0.09	94.43	0.02	0.198	0.002
200	0.068	18.886	5	0.583	0.09	94.43	0.04	0.198	0.002
400	0.068	37.772	3	1.166	0.09	94.43	0.079	0.198	0.002
CASE 2: Varying f ($V_a=90$ m/s, Gas: air)									
f (MHz)									
100	0.068	8.499	1.111	0.262	0.009	9.443	0.018	0.02	0.002
500	0.068	8.499	5.556	0.262	0.045	47.215	0.018	0.099	0.002
1000	0.068	8.499	11.111	0.262	0.09	94.43	0.018	0.198	0.002
CASE 3: Varying gases ($V_a=200$ m/s, $f=1$ GHz)									
Gas									
Hyd	0.16	4.086	5	0.153	0.029	20.431	0.024	0.122	0.006
Air	0.068	18.886	5	0.583	0.09	94.43	0.04	0.198	0.002
Xenon	0.038	62.919	5	1.183	0.111	314.593	0.045	0.224	0.001

TABLE 1. LIST OF DIMENSIONLESS GROUPS FOR THE VARIOUS SIMULATED CASE STUDIES.

Physical Property	Symbol	Value
Beam Width	b	$2\mu\text{m}$
Beam Thickness	t	$0.1\mu\text{m}$
Initial Gap Height	h_o	$1\mu\text{m}$
Excitation Frequency	f	100MHz to 1GHz
Velocity Amplitude	V_a	20 m/s to 1000 m/s
Ambient Temperature	T_o	293 K
Ambient Pressure (Air, Hydrogen, Xenon)	P_o	101.1, 101.5, and 109.2 KPa

TABLE 2. MICRO RF SWITCH GEOMETRY AND GAS FLOW CONDITIONS.

the unsteady inertia term to the diffusion term scales as the product of Re and St .

The results discussed in the following subsections correspond to quasi-steady conditions at times longer than the transients time that arise from the initial conditions. The results are presented in terms of the flow speed in the direction normal to the beam at the symmetry plane near the oscillating beam as a function of time, and the force acting on the beam as a function of time.

Effect of Reynolds Number

The first set of simulations investigates the impact of varying the beam speed amplitude (20 - 400 m/s) with beam oscillation frequency of 1GHz. The corresponding ranges of Reynolds and Mach numbers are $1.9 \leq Re \leq 38$ and $0.058 \leq Ma \leq 1.166$, while $ReSt$ remains constant. The dynamic behavior of the gas velocity near the beam at the symmetry plane, shown in Fig. 5, shows that the gas velocity amplitude approaches the vibrating beam velocity amplitude as Reynolds number increases. This is expected since larger Reynolds number implies an increasing role of the convective component of the inertia term with respect to diffusion. In terms of molecular behavior, as the beam veloc-

ity increases the inertia transfer to molecules (i.e. momentum of convection) becomes more dominant and therefore the gas behavior gets closer to the beam motion governed by τ_{osc} . It is also observed from Fig. 5 that as the beam moves towards the substrate (i.e. positive normalized velocity) at $1 \leq \tau_{norm} \leq 1.25$, the flow is still moving in the opposite direction away from the substrate due to the inertia effect from the beam motion during the previous period (i.e. $0.75 \leq \tau_{norm} \leq 1$). It can also be seen that during the half period when the beam is moving towards the substrate, the flow near the beam reverses the direction of its motion at approximately the same time for all the cases considered. At $\tau = 0.25$, the beam reaches maximum speed half way through the gap, and for $\tau > 0.25$, it starts slowing and the flow in its vicinity reverses its direction and starts moving in the same direction as the beam. At $\tau = 0.5$ when the beam has undergone maximum displacement and the gap size is minimum, the flow in its vicinity, with inertia that is dependent on V_a (and thus Re) is still moving towards the substrate. At $\tau > 0.5$, the beam starts moving away from the substrate, while the flow in its vicinity is observed to still move towards the substrate. At this point, we observe different behaviors among the different cases. By inspecting Fig. 5, it can be observed that for $0.5 < \tau < 0.75$, the flow in the beam vicinity starts slowing down for the case with $Re = 37.772$ while it continues to move towards the substrate for the case with $Re = 1.888$. This can again be attributed to inertia effects; the flow inertia is related to beam inertia in the sense that for larger beam speeds, it imparts larger momentum on the nearby molecules. So as the beam starts moving away from the substrate, how quickly the flow reverses its direction and follows it depends on both the beam speed and the Reynolds number of the flow. For small V_a , the beam slow motion allows the flow (with smaller Re) to take longer time to responds to the beam change in motion direction, whereas for larger V_a , the fast moving beam (away from the substrate) forces the flow to respond more quickly. In summary, the cases with larger Re have asymmetric response of the flow velocity in its vicinity due to strong inertia effects, whereas the response is close to harmonic for the small Re case. Moreover, Fig. 5 shows that the flow reverses its direction during the first half period (rise time) much faster than it does in the second half period (drop time) for flows with high flow inertia (ie. Re). This is due to the fact that during the first half period, the beam is moving towards the substrate while the flow with high Re is still moving in the opposite direction towards the beam itself. This forces the flow to respond much quickly to the beam motion as compared to the second half period where the beam moves away from the substrate while the flow inertia propagates more freely towards the substrate without hitting the beam itself.

In terms of the beam's front face force (solid line), the same inertia effect explained before imposes a high force amplitude during the first half period (collision of a relatively high number of molecules moving away from substrate due to inertia ef-

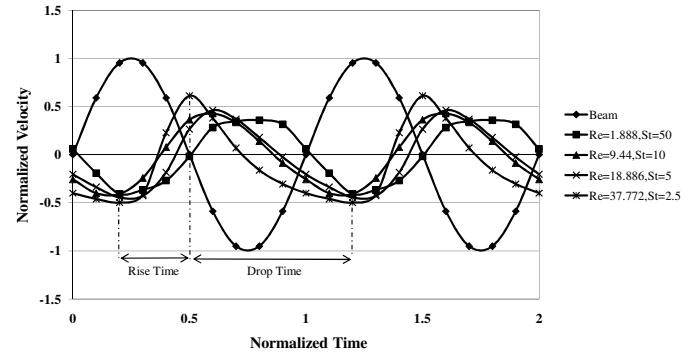


FIGURE 5. FLOW VELOCITY OF GAS FILM AT FIRST CELL NEXT TO THE BEAM WITHIN THE GAS FILM GAP.

fects with the beam moving towards the substrate) as compared to the second half period (molecules keep moving towards the substrate due to inertia effects while the beam is moving away from the substrate). The same phenomena can be observed on the beam's back face force (dashed line) with the difference that during the second half period the flow propagates away from the beam towards the ambient boundary rather than the substrate. Figure 6 also shows that the force amplitude on the beam's front face (facing the substrate) during the first half of the period (as the beam moves towards the substrate) is larger than the force amplitude on the beam's back face during the second half of the period (as the beam moves away from the substrate). This is expected due to the squeeze film effect; the back face is open to the atmosphere whereas the front face senses the higher pressure of the squeeze film in the gap. The amplitude of the net force on the beam (difference between force on front face and force on back face) increases for larger values of Re . As can be seen from Fig. 6, as Re increases, the beam's front face force amplitude increases as seen by points F, F', F'' and F''' . Same trend is observed at the beam's back face force (dashed line) amplitude represented by points B, B', B'' and B''' . This behavior is expected since convection plays a bigger role as Re increases, which is associated with larger momentum transfer to the beam due to collisions with fluid molecules, thus exerting a larger force on the vibrating beam. Further inspection of the force acting on the beam front face (Fig. 6) shows that when the oscillating beam switches its direction of motion (i.e. at $\tau_{norm} = 0.5$), the phase difference between the beam motion and that of the the flow increases as the Reynolds number increases. This is expected since the instant the beam reverses its moving direction the flow with the higher inertia has stronger tendency to keep moving in its direction prior to beam motion direction reversal and thus increasing the phase delay between the beam motion and front plate force.

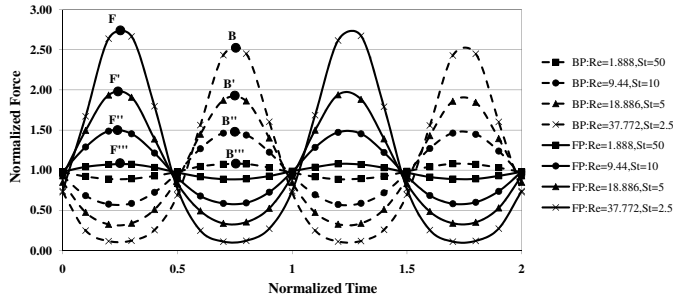


FIGURE 6. FORCES AT BEAM'S FRONT AND BACK FACES.

Effect of Strouhal Number

In the second set of simulations, the beam's speed amplitude is fixed at 90 m/s and the frequency is varied: 100MHz, 500MHz, and 1GHz. Figures 7 and 8 show respectively time evolution of gas velocity at the first cell next to the beam within the film gap and the forces acting on the beam back and front beam face forces. Noting that Reynolds number remains the same, the relative contribution of diffusion with respect to convection does not change. However as the frequency of the plate motion decreases, the product of Re and St decreases, indicating a smaller role of the unsteady component of inertia (compared to diffusion) which brings the flow motion (phase and magnitude) closer to the beam motion, as shown in Fig. 7. Note that since Re is relatively small (i.e. convective component of inertia term plays no dominant role), the rise and drop time are the same and the fluid velocity (near the beam) response is very close to harmonic, differing from the beam motion by amplitude and phase that depend on $ReSt$.

The forces exerted by the gas on the oscillating beam front and back faces are shown in Fig. 8. It is observed that the force amplitude is highest for the beam oscillating with the highest frequency (point F). Interestingly, for the case of beam oscillating at 100MHz, the force on the beam front face experiences a huge drop when compared to the high frequency cases. To better understand this behavior, density contours are plotted at two different time frames within the period; the first (Fig. 9) refers to the normalized time ($\tau_{norm} = 0.3$) at which the beam is moving with maximum velocity towards the fixed substrate while the second (Fig. 10) refers to the normalized time ($\tau_{norm} = 0.8$) at which the beam is moving with maximum velocity away from the fixed substrate. In the first plot (Fig. 9), the high density region is just next to the beam and decreases as one moves towards the substrate. For small beam oscillation frequencies, $ReSt$ is smaller indicating that diffusion plays a larger role when compared to that of the flow inertia and since the ratio of collision to oscillation time scales (Tab. 1) is minimum, this means that molecules have "enough" time to diffuse from the high density region next to the beam to the surrounding low density regions. This effect plays

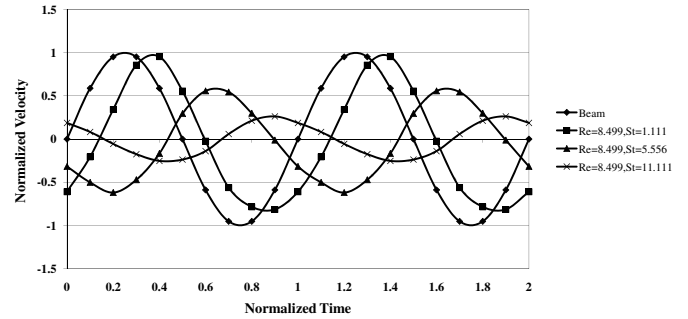


FIGURE 7. FLOW VELOCITY OF GAS FILM AT FIRST CELL NEXT TO THE BEAM WITHIN THE GAS FILM GAP.

a major role in relaxing the pressure at the beam's front face as compared to the cases of 500MHz and 1GHz (points F'' , F' and F respectively in Fig. 8) where the beam oscillates at a much faster frequency and therefore molecules do not have "enough" time to diffuse (i.e. gas entrapment that corresponds to high squeeze number) since the molecular motion is highly affected by the beam oscillation ($(\tau_{coll}/\tau_{osc})_{1GHz} > (\tau_{coll}/\tau_{osc})_{500MHz} > (\tau_{coll}/\tau_{osc})_{100MHz}$). When the beam is moving away from the substrate, (Fig. 10), similar argument can be made: the low density is just next to the beam and increases as we move further towards the substrate. Having a high diffusion rate, molecules have "enough" time to diffuse from the high density region away from the beam to the low density region just next to the beam. This results in higher pressure at the beam's front face as compared to the 500MHz and 1GHz cases (points P'' , P' and P respectively in Fig. 8). As the beam stops and reverses its oscillating direction (i.e. $\tau_{norm} = 0.5$) the relatively high diffusion rate of the 100MHz beam acts against inertia (both Re and $ReSt$ are relatively small) and tends to "fill in" the low density region just next to the beam surface and therefore keeps a higher force (i.e. phase shift) on the beam as compared to the other two high frequency (low diffusion) cases where the force is very close to that exerted by the atmospheric pressure.

Note that at the beam's back face, the force is almost constant (points B'' , B' and B in Fig. 8) due to the fact that the back plate face is open to the atmosphere and therefore no "gas entrapment" effect is experienced by the flow.

Varying Reynolds Number (Constant Strouhal Number)

This set of simulations is carried out for different gases (hydrogen, air, and xenon) while keeping both the beam's speed amplitude fixed (at 200 m/s) and its oscillation frequency (1GHz). Different gases have different effective viscosities, mean free paths, and speeds of sound (and consequently different values of Re , Kn , and Ma). Since Strouhal is constant, as Re decreases,

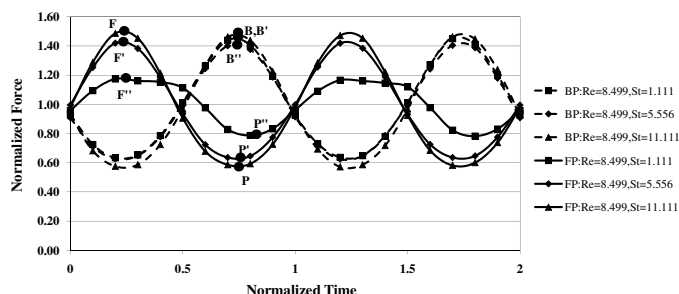


FIGURE 8. FORCES AT BEAM'S FRONT AND BACK FACES.

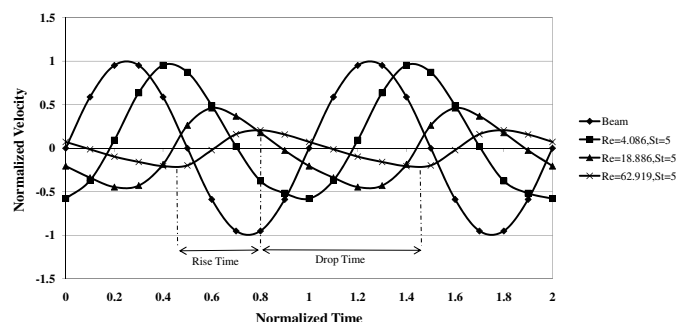


FIGURE 11. FLOW VELOCITY OF GAS FILM AT FIRST CELL NEXT TO THE BEAM WITHIN THE GAS FILM GAP.

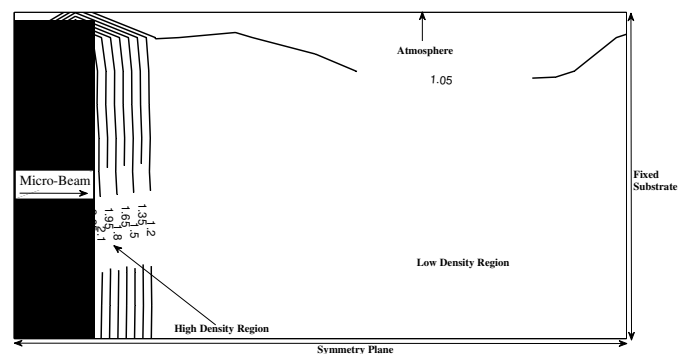


FIGURE 9. DENSITY CONTOURS WITHIN THE FILM GAP THICKNESS AT $\tau_{norm} = 0.3$.

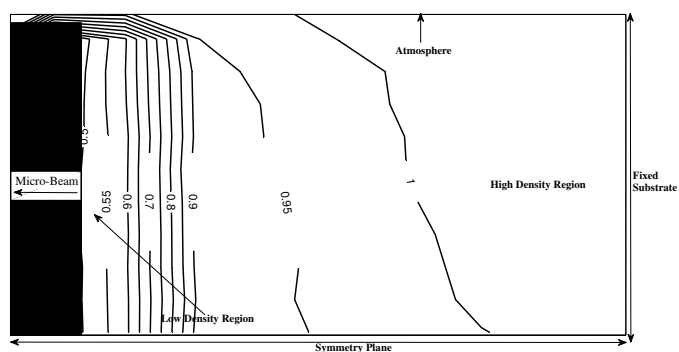


FIGURE 10. DENSITY CONTOURS WITHIN THE FILM GAP THICKNESS AT $\tau_{norm} = 0.8$.

the diffusion term plays an increasing role in governing the flow motion and simultaneously the product of Re and St decreases forcing the flow velocity (phase and magnitude) to get closer to the vibrating beam motion. In terms of molecular behavior, by looking at the ratio of the collision time scale to the convection and diffusion time scales (Tab. 1), it is found that the molecules of the heavy gas (i.e. Xenon) are affected more by

convection whereas the light gas (i.e. Hydrogen) is affected more by diffusion. For the diffusion term, this is expected since in terms of molecular behavior, Hydrogen is the lightest of the three gases and therefore it has the highest diffusion rate since light molecules diffuse faster than the heavier ones. For the convection term, molecular mass plays a major role in inertia transfer and therefore the heaviest gas (Xenon) has the highest convection term. However, by looking at the ratio of the convection and diffusion time scales to the oscillatory time scale, it is found that the decisive term is the latter (the product of Re and St) since the former has a constant value (St number is fixed). Therefore, Hydrogen gas velocity is expected to be closest to the beam behavior and this is shown in Fig. 11. Using similar arguments to those employed in analysis of the first simulation set, the heavy gas flow with high inertia effects (i.e. high Re) has a flow rise time that is much faster than the drop time as depicted on Fig. 11. Consequently, this higher inertia effect departs the flow velocity profile from the sinusoidal harmonic response especially during the flow drop time.

The same inertia effect explained in the first simulation set shows a high force amplitude during the first half period relative to the second half period for each of the beam's front face force (solid line) and back face force (dashed line) for high inertia flows as depicted on Fig. 12. For the beam's front and back face forces, the heavier gas exert more pressure on the beam (points F, F' and F'' in Fig. 12) due to its larger molecular mass since the pressure a molecule exerts on the oscillating beam is proportional to the molecule's rate of change of momentum. The beam is oscillating at 1GHz for all three gas mediums and therefore the "gas entrapment" phenomena prevails as discussed in Case 2 leading to no significant difference between the back and front plate forces ($F \sim B, F' \sim B', F'' \sim B''$ in Fig. 12). Figure 12 shows that as Re and $ReSt$ increase (i.e. moving from the light to the heavy gases), the inertia term plays a more dominant role (due to larger molecular mass as opposed to larger speed encountered in the first simulation set) with respect to the diffusion term which tends to increase the delay at $\tau_{norm} = 0.5$.

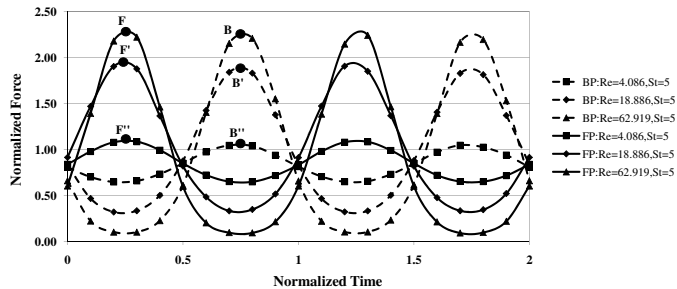


FIGURE 12. FORCES AT BEAM'S FRONT AND BACK FACES.

CONCLUSION

The dynamic force on a micro RF switch oscillating above a substrate imparted by a surrounding has medium is investigated for a range of oscillating speed amplitude and frequency, and for three gases of different molecular mass. The micro flow behavior of the gas film between the beam and the substrate and its relation to the forces acting on the front and back sides of the beam are interpreted in terms of the relative contributions of diffusion and the unsteady and convective components of inertia, as quantified by the various dimensionless groups. It has been observed that for larger values of Reynolds number, the force on the side of the beam facing the substrate (front face) departs from the harmonic profile of the beam motion imposed, the flow velocity rise time decreases with respect to drop time, the force amplitude at beam front face during the first half period (i.e. when beam is moving towards substrate) is larger than the beam back face force during the second half period (beam moving away from substrate) due to squeeze damping effects, and the force amplitude on both beam faces during the first half period is larger than their counterparts during the second half period due to inertia effects. For small Strouhal number, the phase difference between the beam and flow within the gap decreases. When Re and St are small, diffusion plays a dominant role over inertia, increasing the phase difference between the beam's motion and front face force and decreasing the front face force amplitude. This corresponds to relatively low squeeze number where the gas has enough time to diffuse and escape from getting trapped between the oscillating beam and the fixed substrate.

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